

Abraham Morales Rodríguez

Software Developer | DevOps & Elearning Systems

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Professional Profile

Software developer with hands-on production experience in **CI/CD pipeline design**, containerized deployments, and **microservices architecture**. Led the **technical integration** of a Retrieval-Augmented Generation (RAG) system with **Moodle**, and I am **technical lead** for a core module of the open-source LMS **ILIAS**.

I work daily with **Docker** and **Linux system administration** on the **DevOps** side, **TypeScript/React/Next.js** on the frontend, and **PHP and Java/Spring** on the backend. I am passionate about **e-learning** and how technology and well-engineered infrastructure can support the learning process.

Experience

Surlabs

Jan. 2025 - Present

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- Led the **technical integration of Moodle** with SURContext, Surlabs' Retrieval-Augmented Generation (RAG) system developing the **Moodle plugins** that connect the **LMS directly to the RAG pipeline**, and leading the technical implementation end to end.
- **Technical lead** for Survey (PHP), a core module of the open-source Learning Management System (**ILIAS**), owning its technical evolution, defining priorities, managing incidents, and coordinating requirements with clients. Took over the module's development, **resolving 40% of the accumulated backlog** to drive it forward.
- Designed and implemented a **version-control protocol** based on **GitHub Actions** and releases, defining a **CI/CD** flow with separate development, testing, and production environments. The change **reduced deployment-related** incidents and enabled more stable, predictable releases.
- Developed **new features**, shipped **improvements**, and resolved **incidents** in custom and opensource plugins for the **ILIAS** LMS platform, managing the **full technical ticket lifecycle**.
- Audited and built **load tests** using **Locust** on a RAG system to evaluate its capacity and concurrency, helping identify improvement points and inform **infrastructure** decisions for each client.

Software developer

- Built ***Checkeapp*** (**TypeScript, Next.js, React**), a real-time fact-checking platform using **Retrieval-Augmented Generation (RAG)** to pull live web sources and generate **verifiable**, **reference-backed reports**, reaching **95% accuracy**. Built under ***ILENIA*** & ***ALIA***, the Spanish government's public AI initiative, in collaboration with the *University of the Basque Country*.
- Built an internal business-management platform using **Express**, building **RESTful APIs** and integrating **PostgreSQL** for data management. The platform focused on service integration and process automation, improving time spent managing tasks by **30%.**

Projects

OpoFlash

Jun. 2024 - Present

app.opo-flash.com

- Designed and built a **microservices-based** e-learning system integrated with a custom **RAG** system that **analyzes student-uploaded** content and **automatically generates flashcards and summaries** contextualized to reinforce learning.
- Implemented a **spaced-repetition engine** in Python (**FSRS-5**) that adapts review intervals to each student's performance, optimizing long-term retention, based on the forgetting curve.

Technical Skills

Linux & DevOps: Linux server administration, Docker, Docker Compose, CI/CD (GitHub Actions), Coolify, Plesk, performance/load testing (Locust).

Cloud: AWS (EC2, S3, RDS, IAM, VPC, Bedrock).

LMS: Moodle (plugin development, RAG integration) & ILIAS (plugin development, system administration and e-learning management).

Backend: PHP, Java (Spring Boot), C# (.NET), Node.js (Express), Python (Django).

Frontend: React, Next.js, Vite, TypeScript, Chakra UI, Tailwind CSS, Shadcn/ui, Framer Motion, Responsive Design, Accessibility.

Database: Supabase (PostgreSQL, Auth), MySQL, PostgreSQL, MongoDB.

Testing: Jest, Vitest, Locust

Education

Higher Technical Degree in Computer Science (Web Application Development).
IES Hermanos Machado — *2023 - 2025*

Cambridge English: C1 Advanced (CAE) — *2025*